

Database systems & Information integration

Hands on Assignment 1:

Exploring the Sakila Database

Lecturer

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Contents

1	Instructions	3
2	Part A — Describe Statements	5
3	Part B — Basic WHERE Queries	5
4	Part C — Distinct	5
5	Part D — Combining Conditions	5
6	Part E — NOT / Negation	5
7	Part F — Leveling Up (Still No Joins)	6

1 Instructions

Step 1: Import Sakila Database via Administration Tab

1. Open MySQL Workbench and connect to your local server.
2. Click the **Administration** tab on the left sidebar.
3. Under **Data Import / Restore**, click **Data Import / Restore**.
4. Choose **Import from Self-Contained File** and select `sakila-schema.sql`.
5. Under **Default Target Schema**, select an existing schema or check **New** to create `sakila`.
6. Click **Start Import** to create the tables.
7. Repeat the process with `sakila-data.sql` to insert sample data.
8. Verify import by right-clicking **Schemas** → **Refresh All**.

Step 2: View ER Diagram of the Database

1. Go to **Database** → **Reverse Engineer**.
2. Select your connection and click **Next**.
3. Select the `sakila` schema and click **Next**.
4. Follow the prompts: **Next** → **Finish**.
5. The EER diagram will open showing tables, primary keys, foreign keys, and relationships.

Instructions

You must answer the following questions using only:

- SELECT, FROM, WHERE
- AND, OR, NOT, BETWEEN, IN
- Comparison operators: =, !=, >, <, >=, <=
- DISTINCT, ORDER BY, LIMIT
- Simple aggregates from a single table: COUNT, MAX, MIN, AVG
- Subqueries (only inside WHERE).

Strict Rule: Do NOT use JOINS. Only single-table queries or subqueries allowed.

2 Part A — Describe Statements

1. Describe the `film` table.
2. Describe the `customer` table.

3 Part B — Basic WHERE Queries

3. List the titles of all films with rental duration greater than 5.
4. Show all actors whose first name is exactly “NICK”.
5. Retrieve all customers who are not active.
6. List all films with length between 80 and 100 minutes.
7. Show films with rental rate less than 1.00.
8. List all films that are rated “PG” or “G”.
9. Show films with replacement cost greater than 20.00.

4 Part C — Distinct

10. Show all distinct film ratings.
11. Show all distinct `language_id` values in the `film` table.

5 Part D — Combining Conditions

12. List films with rental rate = 2.99 **and** rental duration greater than 4.
13. List customers in store 1 who are active.
14. Show actors whose first name is “PENELOPE” **or** last name is “MONROE”.

6 Part E — NOT / Negation

15. Show all films that are **not** rated “R”.
16. Show all customers whose first name is not “MARY”.
17. List all films whose length is **not** between 40 and 120.

7 Part F — Leveling Up (Still No Joins)

18. Show the top 10 longest films (order by length descending).
19. Find the film with the maximum replacement cost.
20. Find the film with the minimum rental duration.
21. List the 5 cheapest films (by rental_rate).
22. Show all films whose length is greater than the **average** film length (use a subquery).
23. Show all films with rental_rate **greater** than the **average rental_rate**.
24. List all actors whose actor_id appears in the top 10 highest actor IDs:

Hint: Use *actor_id* IN (subquery with ORDER BY ... LIMIT 10).

25. Show all customers whose customer_id is **not** in the list of the first 50 customers.
26. Count how many films have rating “R”.
27. Count how many customers are active.
28. Show films whose film_id is in the set of the 10 most expensive films (by replacement cost).
29. Show films whose length is **greater** than the maximum length of any film rated “PG”.
30. Show all films whose rental_duration is **less** than the minimum rental_duration of “G”-rated films.

8 Answer Key

Part A

```
DESCRIBE film;  
DESCRIBE customer;
```

Part B

```
SELECT title FROM film WHERE rental_duration > 5;  
SELECT first_name, last_name FROM actor WHERE first_name = 'NICK';  
SELECT * FROM customer WHERE active = 0;  
SELECT film_id, title, length FROM film WHERE length BETWEEN 80 AND 100;  
SELECT title, rental_rate FROM film WHERE rental_rate < 1.00;  
SELECT title FROM film WHERE rating = 'PG' OR rating = 'G';  
SELECT title, replacement_cost FROM film WHERE replacement_cost > 20.00;
```

Part C

```
SELECT DISTINCT rating FROM film;  
SELECT DISTINCT language_id FROM film;
```

Part D

```
SELECT title FROM film WHERE rental_rate = 2.99 AND rental_duration > 4;  
SELECT * FROM customer WHERE store_id = 1 AND active = 1;  
SELECT * FROM actor WHERE first_name='PENELOPE' OR last_name='MONROE';
```

Part E

```
SELECT title FROM film WHERE rating != 'R';  
SELECT * FROM customer WHERE first_name != 'MARY';  
SELECT * FROM film WHERE length NOT BETWEEN 40 AND 120;
```

Part F — Harder Questions

```
SELECT title FROM film ORDER BY length DESC LIMIT 10;  
  
SELECT title FROM film  
WHERE replacement_cost = (SELECT MAX(replacement_cost) FROM film);  
  
SELECT title FROM film  
WHERE rental_duration = (SELECT MIN(rental_duration) FROM film);  
  
SELECT title FROM film ORDER BY rental_rate ASC LIMIT 5;  
  
SELECT title FROM film  
WHERE length > (SELECT AVG(length) FROM film);
```

```
SELECT title FROM film
WHERE rental_rate > (SELECT AVG(rental_rate) FROM film);
```

```
SELECT * FROM actor
WHERE actor_id IN (SELECT actor_id FROM actor ORDER BY actor_id DESC LIMIT 10);
```

```
SELECT * FROM customer
WHERE customer_id NOT IN (SELECT customer_id FROM customer ORDER BY customer_id LIMIT 10);
```

```
SELECT COUNT(*) FROM film WHERE rating='R';
SELECT COUNT(*) FROM customer WHERE active=1;
```

```
SELECT * FROM film
WHERE film_id IN (
SELECT film_id FROM film ORDER BY replacement_cost DESC LIMIT 10
);
```

```
SELECT * FROM film
WHERE length > (
SELECT MAX(length) FROM film WHERE rating='PG'
);
```

```
SELECT * FROM film
WHERE rental_duration < (
SELECT MIN(rental_duration) FROM film WHERE rating='G'
);
```